Giving Permanent Permission to Pico Device

Demo Video Link:

https://www.loom.com/share/5d0454c0f57341b7a05840d0aa65a7b3?sid=e8c1bcc4-8abd-46c5-a69e-a847310f8250

When the serial port is opened, permission is required.

Follow these steps to give permanent permission to the Pico device.

Step# 1: Identify the Device

Plug in your device and run the following command:

```
ls -l /dev/ttyACM0
```

Step# 2: Check Device Attributes

Get attributes like idVendor and idProduct by running:

```
udevadm info -a -n /dev/ttyACM0
```

Look for the first block with idVendor and idProduct attributes.

Step# 3: Create a udev Rule

Create a new udev rule file:

```
sudo nano /etc/udev/rules.d/99-usb-serial.rules
```

Add the following line (replace with your device's idVendor and idProduct values):

```
ATTRS{idVendor}=="1234", ATTRS{idProduct}=="5678", MODE="0666"
```

Step# 4: Reload udev Rules

Reload udev rules and reconnect the device:

```
sudo udevadm control --reload-rules
sudo udevadm trigger
```

Step# 5: Verify

Check the permissions:

```
ls -l /dev/ttyACM0
```

Commands to Run Before Building

Demo Video Link:

https://www.loom.com/share/b4081a4775e1481c90bb6160e8da9d65?sid=e43c817b-b9b0-4ef1-9929-10c3a4cc06ad

Before starting the build process, ensure the necessary packages and tools are installed by running the following commands:

Update package list:

```
sudo apt update
```

Install essential tools:

```
sudo apt install cmake
```

```
sudo apt install build-essential
```

Install Qt5 development libraries:

```
sudo apt install qt5-qmake qtbase5-dev qtchooser qtbase5-dev-tools
```

```
libqt5webkit5-dev libqt5svg5-dev
```

Install Google Test and compile it:

```
sudo apt-get install libgtest-dev
```

```
sudo apt-get install cmake
```

```
cd /usr/src/gtest
```

```
sudo cmake .
```

```
sudo make
```

```
sudo cp libgtest*.a /usr/lib
```

Install additional libraries:

```
sudo apt install libboost-all-dev libssl-dev libprotobuf-dev protobuf-compiler
```

Install Git:

```
sudo apt install git
```

Install C++ compilers:

```
sudo apt install g++ gcc
```